

AKASHDIP MAHAPATRA

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PROFESSIONAL SUMMARY

Results-driven Site Reliability Engineer with a strong foundation in mechanical systems thinking, specializing in **infrastructure automation, CI/CD pipelines and distributed systems**. Proven ability to **accelerate development velocity** and reduce operational toil through Python scripting, design highly reliable AWS cloud architectures, and optimize complex enterprise applications.

CORE COMPETENCIES

- **Security & Observability: (SRE):** Datadog, CloudWatch, Vulnerability Management (CVE patching), SLA/SLO tracking, incident management.
- **DevOps & CI/CD:** GitHub Actions (utilizing IAM Role ARNs via OIDC for passwordless authentication), Jenkins, Terraform (Infrastructure as Code), Advanced Git Version Control (resolving complex remote history mismatches, local vs. remote state reconciliation).
- **Programming & Scripting:** Python (Boto3, Data Structures & Algorithms), Bash, C, SQL, **AI-Assisted Development (Leveraging GenAI for script automation, code optimization, and RCA)**.
- **Cloud Infrastructure:** AWS (Lambda, ECR, ECS, S3, IAM, VPC, EventBridge, **Apache Kafka** (Amazon MSK), EC2, DynamoDB), & fundamentals of Azure (Azure DevOps) and Google Cloud Platform.
- **Container Orchestration & Cloud:** Docker, AWS ECS, ECR, AWS Serverless (Lambda, EventBridge), Apache Kafka (Amazon MSK).
- **Operating Systems & Networking:** Linux (Ubuntu, RHEL), TCP/IP, DNS, HTTP/HTTPS, Nginx, Wireshark, nmap.

PROFESSIONAL EXPERIENCE

1. **Tata Consultancy Services (TCS)** | Onsite *Data Engineer (DevOps & Infrastructure Focus under Global Aviation Client)* | Nov 2025 - Present
 - Provision, manage, and scale multi-tiered AWS cloud infrastructure utilizing Terraform (IaC) and Docker, actively managing 50+ distributed compute nodes, Amazon MSK/Kafka brokers, and serverless execution environments across multiple production tiers.
 - Manage both secure internal event-driven pipelines and public-facing APIs, ensuring reliable data delivery for enterprise clients, global aviation personnel, and downstream B2B applications processing 200 GB to 500 GB of IoT data daily.
 - Lead Platform Level-2 (L2) shift support and incident response, actively monitoring 60+ Datadog infrastructure alerts across 5 environments to enforce strict SLAs, support performance, and maintain near-zero deployment downtime.
 - Execute advanced production troubleshooting via SSH, utilizing deep Linux OS administration and log extraction techniques to rapidly resolve live infrastructure anomalies, complex routing issues, and DNS failures.
2. **Tata Consultancy Services (TCS)** | Remote *Network & DevOps Trainee (Global Technology Client)* | Oct 2025
 - Troubleshoot deep Linux OS and networking issues, analyzing **system calls (strace)**, **inodes**, and network traffic using TCP dump and Wireshark.
 - Deployed and configured Nginx reverse proxies and load balancers, successfully resolving HTTP 403/404 errors and DNS routing failures.
 - Hardened server security by implementing strict file permissions, writing custom Bash scripts for system automation, and configuring network firewalls (UFW, iptables, AWS Security Groups).

SYSTEMS ENGINEERING & AUTOMATION PROJECTS

ENTERPRISE ENGINEERING PROJECTS (Tata Consultancy Services)

1. Post-Deployment Infrastructure Validation Engine

- Engineered a Python and Boto3 engine from scratch to auto-discover AWS resources (Lambda, SNS, API Gateway) and automate a 30-minute manual deployment checklist into a 2-minute zero-error pipeline.
- Integrated CloudWatch metrics and log filtering to enforce strict baseline health checks, generating automated, audit-ready performance reports for a 30-person DevOps team.

2. UAT Data Pipeline & Health-Check Automation

- Built a dedicated QA testing harness and Python automation script to validate 1:1 payload comparisons and secure SHA-256 hashing across multi-tiered UAT environments.
- Automated manual JSON comparison processes and tracked DLQ routing via CloudWatch Logs Insights, accelerating QA regression testing and ensuring downstream data reliability.

3. High-Throughput Data Extraction (Automated Manual Dumps)

- Developed Python-based software solutions utilizing Boto3 to eliminate manual extraction processes and automate complex data pipeline validation workflows.
- Orchestrated high-throughput data extraction bridging legacy Oracle SQL databases with AWS S3 cloud storage, ensuring reliable, performant data ingestion for downstream analytics teams.

4. DevSecOps & Container Vulnerability Remediation

- Spearheaded comprehensive DevSecOps remediations to patch critical CVEs in containerized Python Lambdas via dependency upgrades.
- Implemented multi-stage Docker builds to restructure Dockerfiles, successfully hardening container security, reducing AWS Lambda cold-start overhead, and achieving a 42% image size reduction (607MB to 350MB).

ACADEMIC & INDEPENDENT PROJECTS

1. Lightweight Python Web Server (OSI Layer 4/7)

- Bypassed standard web frameworks to build a raw HTTP server from scratch using basic Python TCP Socket Programming, handling low-level GET/POST request parsing.

2. Network Security & Packet Forensics

- Engineered a Python-based wireless penetration testing system to scan 802.11 traffic, capture handshakes, and test access point vulnerabilities.

PUBLICATIONS & APPLIED RESEARCH (Data Science & Machine Learning)

Modeling and Optimization of Surface Roughness of Electrodeposited Nickel Coating using Taguchi and Bonobo Optimizer | Jadavpur University Press (ISBN: 978-81-993635-8-8)

- Applied the Bonobo Optimizer (a meta-heuristic optimization algorithm) and Taguchi design of experiments to model, predict, and optimize complex physical processes.
- Demonstrated advanced capability in data-driven mathematical modeling, statistical analysis, and implementing optimization algorithms to solve multi-variable engineering challenges.

EDUCATION

Academy of Technology Engineering College | Hooghly, West Bengal Bachelor of Technology in Mechanical Engineering | Sep 2021 - 2025

- Applied mechanical systems-design principles and analytical problem-solving skills to complex cloud infrastructure and software architecture.
- Mechanical Capabilities & Projects:** AutoCAD, SolidWorks (Advanced CAD, Sheet-Metal, Weldments), Ansys, MATLAB | **Key Focus:** Thermodynamics, Solid/Fluid Mechanics | **Applied Work:** 3D-printed working drone, fully functional V8 engine design, published academic research.

CERTIFICATIONS & QUALIFICATIONS

- AWS Certified Cloud Practitioner (**Perfect Score: 1000/1000**)
- Microsoft Certified: Azure Network Engineer Associate
- Google Cloud Certified Professional
- Oracle Cloud Infrastructure 2024 Certified Foundations Associate
- GitHub Certifications: Advanced Security, Actions, and Administration
- Postman Student Expert
- NASA Open Science 101 Qualification.